

	
20	A Since the thermistor's resistance decreases as its temperature increases, in accordance with the potential divider principle, the potential difference (p.d.) across it decreases with increasing temperature. The thermistor and the fixed resistor are in series thus their potential differences add up to 10 V. Therefore the p.d. across the fixed resistor increases as the p.d. across the thermistor decreases due to increase in temperature. When the p.d. across the fixed resistance exceeds the built-in p.d. of the diode (0.3 – 0.7 V) the diode becomes forward biased and current starts to flow downwards through the lamp to light it up. Option C is wrong as the diode is always in reverse bias hence the lamp will not light up as current will not flow through the diode even if the p.d. in the fixed resistor exceeds the diode's built-in potential difference.

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